

HOSPICART

Summary of the System's Intent and Features



Databases 2025

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1. System Intent

“HospiCart” is a Java-based ordering platform tailored for hospitals and healthcare professionals. Inspired by the Amazon model, it simplifies the procurement of medical products through a role-based interface that supports doctors, nurses, and suppliers. The goal is to digitize and centralize medical supply management in clinical settings, improving efficiency and reducing manual workloads.

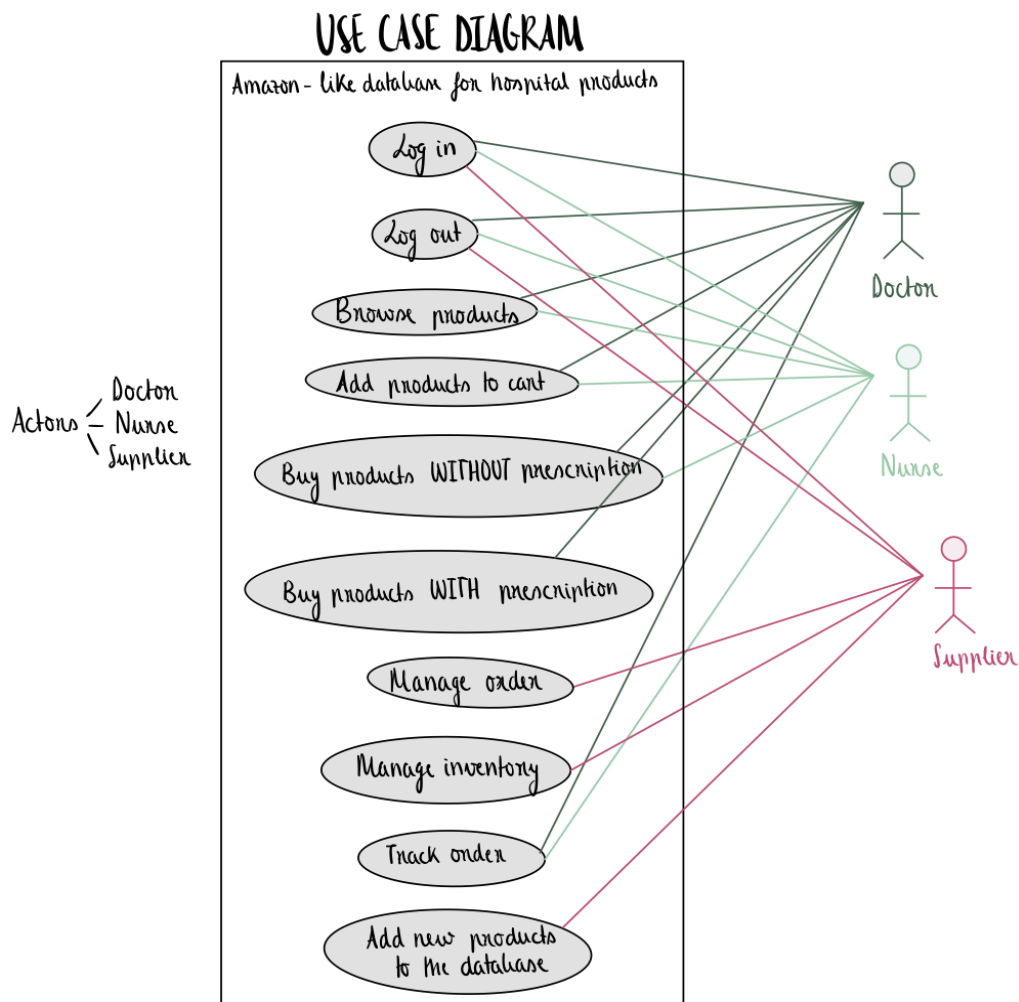
2. Main Features

- Intuitive, role-based interfaces for:
 - **Doctors:** browse products, place orders, manage cart and personal information.
 - **Nurses:** similar to doctors but can create orders of products that do not require prescription.
 - **Suppliers:** manage products, stock levels, and view orders.
- Centralized product catalog and order system.
- Secure login, registration, and role assignment.
- Encrypted password management (via JPA).
- Layered architecture: POJOs, Interfaces, JDBC, JPA, Application Logic, Exceptions.
- Dual database access via **JDBC** and **JPA**.
- Custom exception management.
- Console-based UI.

3. Supporting Documents Developed

To support and justify the system architecture and functionalities, the following documents were created:

- **Use Case Diagram:** definition and visualization of system-user interactions.



- **Use Cases:**

Name	Browse Products
Preconditions	The doctor must be logged into the system.

Postconditions	The doctor views product categories and details. The doctor may choose to add products to the cart.
Primary Actors	Doctor, Nurse
Main Scenario	1. The user navigates to the product catalog. 2. The user searches for specific items or browses by category. 3. System displays relevant product details.

Name	Buy Sanitary Products Without Prescription
Preconditions	The user is signed in to HospiCart.
Postconditions	The payment of the order was accepted. A track number is generated for the order.
Primary Actors	Nurse, Doctor.
Main Scenario	1. The user logs in to HospiCart. 2. The user browses the catalog of HospiCart and adds the desired items to the shopping cart. 3. The user pays the products he/she added to the cart and the payment is accepted. 4. The system creates the order record. 5. The user receives confirmation of the successful operation alongside the tracking number of the order.

Name	Buy Sanitary Products With Prescription
Preconditions	The user is signed in to HospiCart. Verify that the user is a doctor.
Postconditions	The payment of the order was accepted. A track number is generated for the order.
Primary Actors	Doctor.
	1. The doctor logs in to HospiCart. 2. The doctor browses the catalog of HospiCart and adds the desired

Main Scenario	items to the shopping cart. 3. The doctor pays the products he/she added to the cart and the payment is accepted. 4. The system creates the order record. 5. The doctor receives confirmation of the successful operation alongside the tracking number of the order.
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Name	Track order
Preconditions	The user is signed in to HospiCart. The user has introduced a tracking number. The system has checked whether the tracking number is valid (if it is associated with an order).
Postconditions	The user is able to view the state and information of his/her order.
Primary Actors	Nurse, Doctor.
Main Scenario	1. The user logs in to HospiCart. 2. The user introduces the track number of the order he/she wants to track. 3. The application retrieves the information of the order with the introduced tracking number.

Name	Log in
Preconditions	The user must be registered into the system.
Postconditions	The user gains access to the system administration functions.
Primary Actors	Supplier, Doctor, Nurse
Main Scenario	1. The user accesses the login page. 2. He enters his username and password. 3. The system validates the credentials. 4. If the credentials are correct, the system redirects the user to the administration panel. 5. If the credentials are incorrect, an error message is displayed.

Name	Log out
Preconditions	The user must be logged into the system.
Postconditions	The session is successfully closed.
Primary Actors	Supplier, Doctor, Nurse
Main Scenario	1. The user accesses the “Logout” option. 2. The system logs out and redirects to the home page. 3. A message is displayed confirming the logout.

Name	Product Registration
Preconditions	The administrator must be logged into the system. The system checks if the product is not already added.
Postconditions	A new product is added to the medical devices catalog.
Primary Actors	Supplier
Main Scenario	1. The supplier accesses the “Register Products” section. 2. He/she enters the new product data (name, description, price, stock, category). 3. The system validates the data entered. 4. If the data are correct, the system saves the product in the database. 5. A confirmation message is displayed.

Name	Inventory Management
Preconditions	The administrator must be logged into the system. There must be an inventory of products in the database.
Postconditions	The inventory is updated with the changes made (adding, modifying or deleting products).

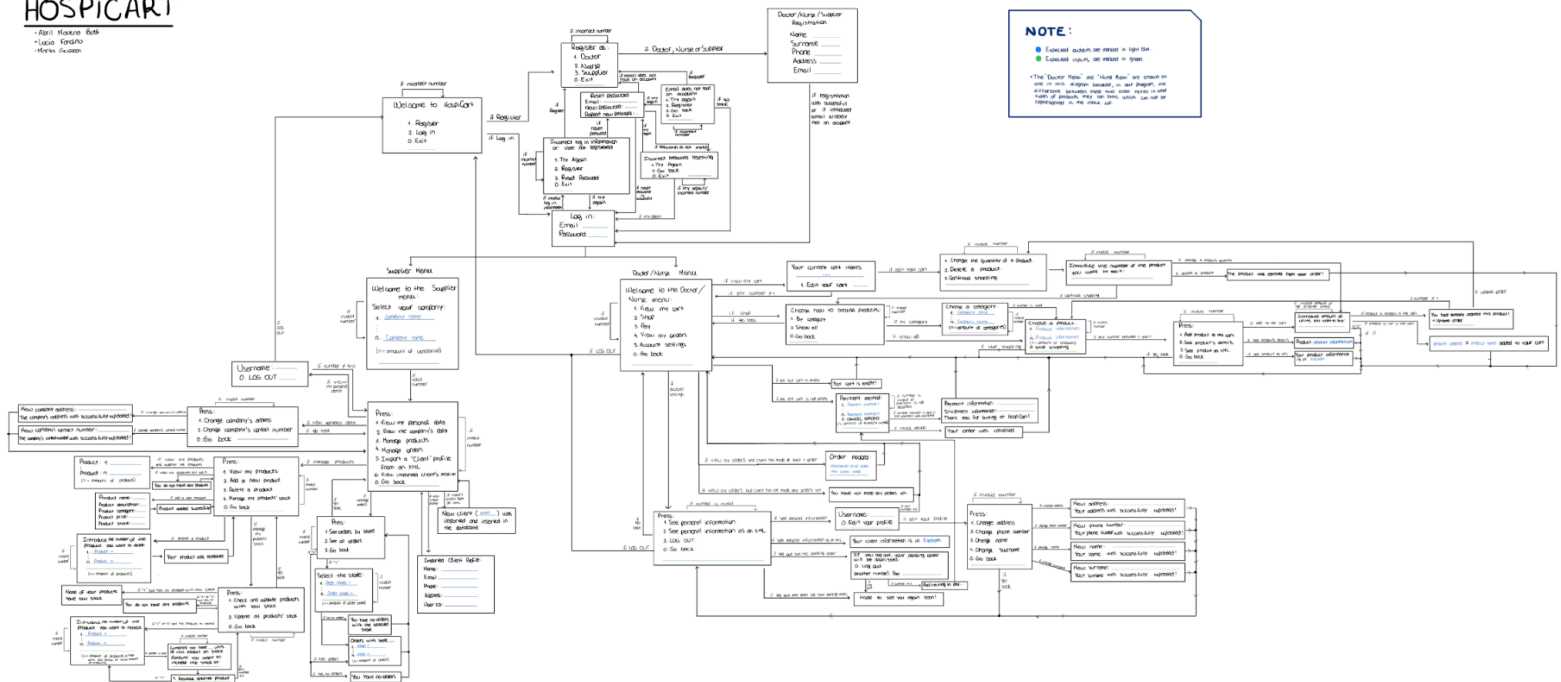
Primary Actors	Supplier
Main Scenario	1. The supplier accesses the “Inventory Management” section. 2. The system displays the list of current products with their stock. 3. The supplier chooses one of the following actions: - Add a new product (entering name, description, price, stock, etc.). - Modify an existing product (changing its price, stock, etc.). - Remove a product from the inventory. 4. The system updates the database and displays a confirmation message.

- **Traceability Matrix:** links between functional requirements and implementation elements.

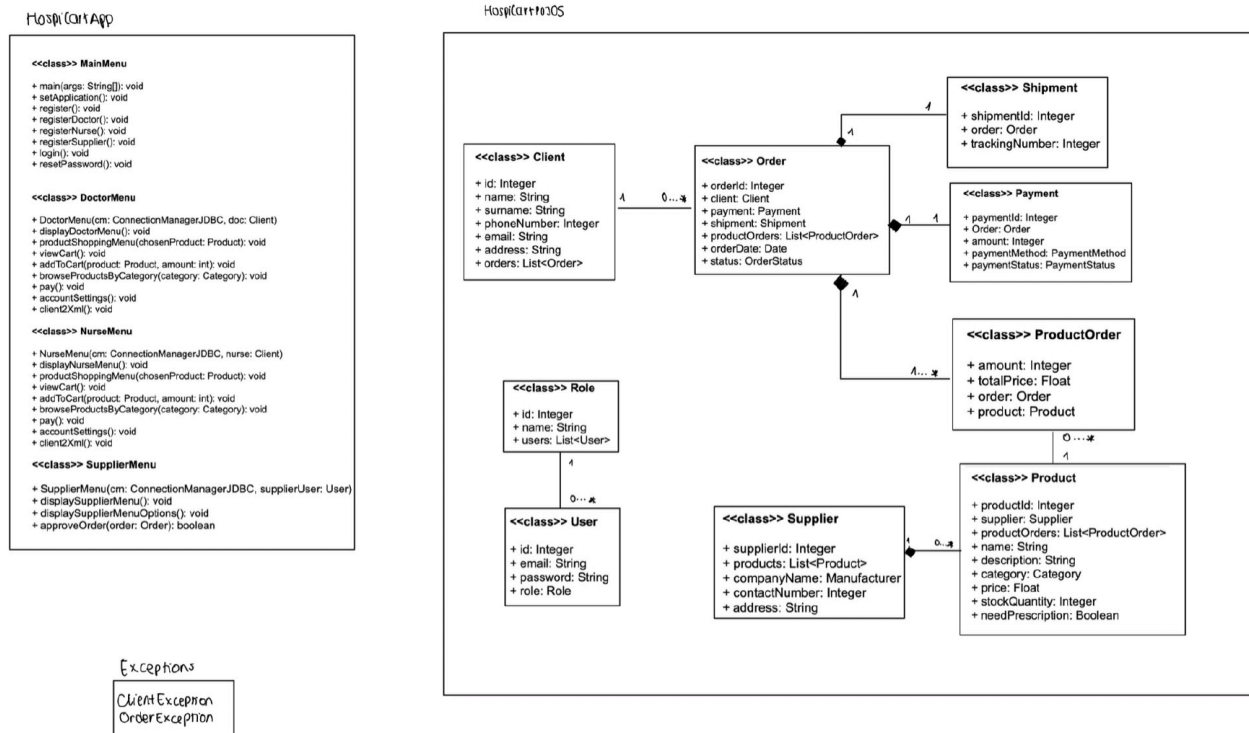
HOSPICART		Req - The introduced email does not have an account	Req - The user has a registered account	Req - The user has logged in	Req - The username is valid	Req - The password is valid	Req - The user has prescription	Req - The cart is not empty	Req - The payment is accepted	Req - A tracking number is generated	Req - The user has at least one order associated to his account	Req - The user is not trying to add a repeated product	Req - Products with no stock can not be bought	Req - Restock products
UC-1 Register		✓			✓	✓								
UC-2 Log In			✓		✓	✓								
UC-3 Log out			✓	✓										
UC-4 Search products			✓	✓										
UC-5 Add products to cart			✓	✓									✓	
UC-6 Make purchase with prescription			✓	✓			✓	✓	✓	✓				
UC-7 Make purchase without prescription			✓	✓				✓	✓	✓				
UC-8 Track order			✓	✓							✓			
UC-9 Manage inventory			✓	✓								✓		✓
UC-10 Register products			✓	✓										

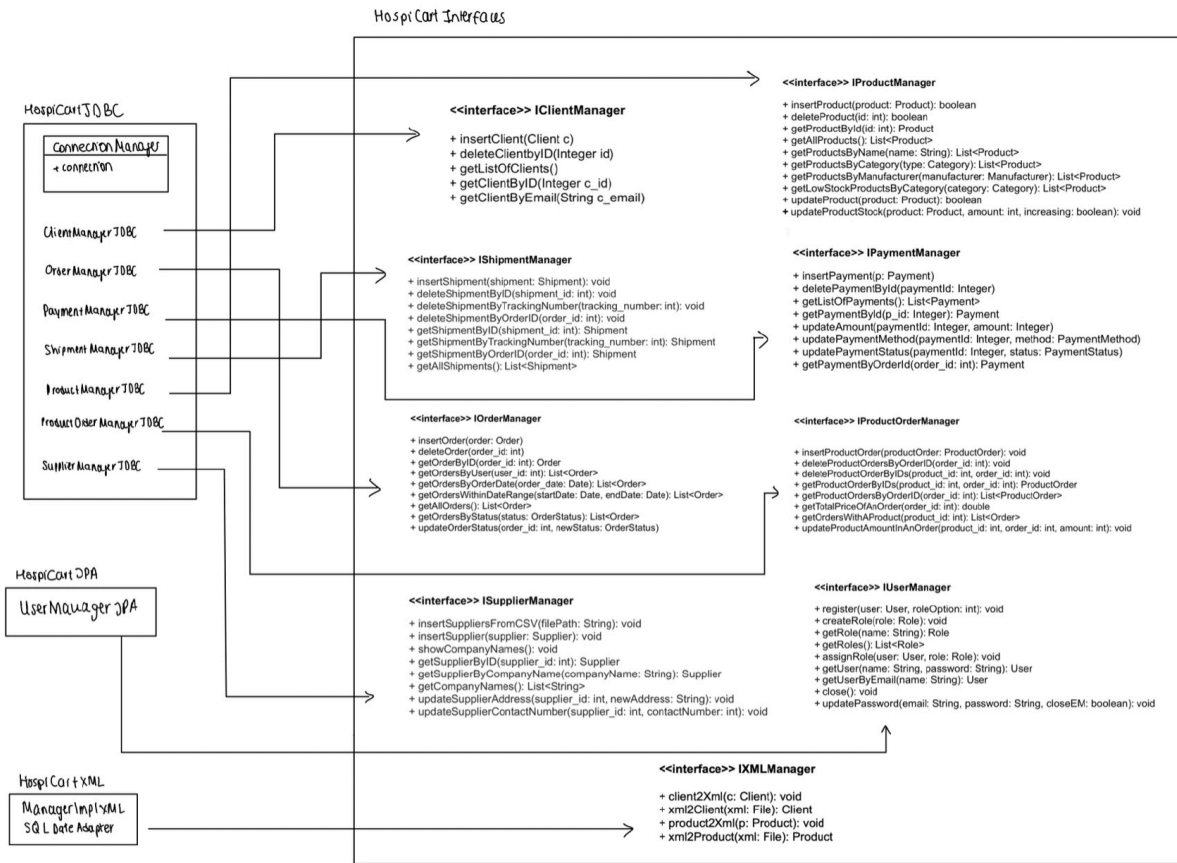
- **Mockup:** interface sketches.

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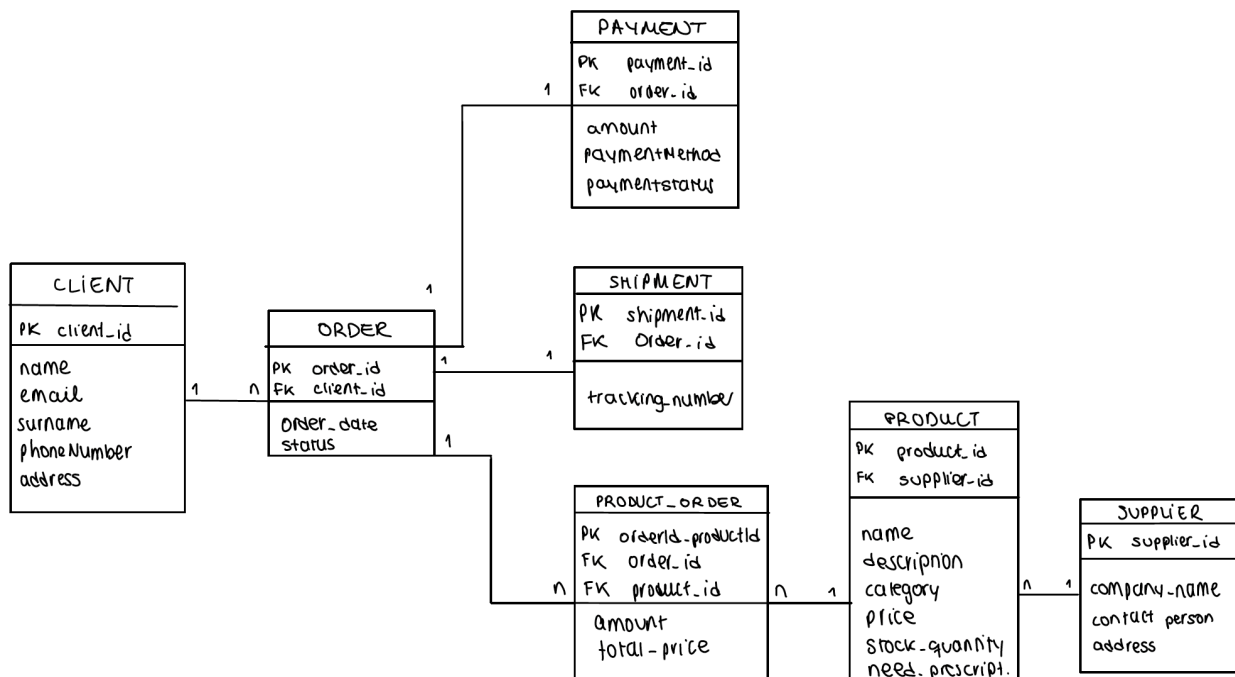


- **UML Class Diagram:** details the object-oriented structure of the system.





- **Entity-Relationship (ER) Diagram:** models the logical database design.



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- **User Manual:** step-by-step guidance for each role (doctor, nurse and supplier).
 - **Design Document:** technical explanation of design decisions, technologies, and structure.

4. Extra Development Features Implemented

The project also includes the following enhancements beyond the standard requirements:

1. **Written Design Document**
Fully detailed technical design report covering architectural choices and structural reasoning.
2. **Password Encryption**
All user passwords are securely encrypted via JPA mechanisms to ensure data protection.
3. **User Manual**
A clear and comprehensive manual helps users of all roles interact efficiently with the system.
4. **JUnit Testing of POJO Methods**
All POJOs are validated through unit testing to guarantee proper behavior and integrity.